

# Notebook 07: Panel Regressions & Panel VAR

This notebook implements the core econometric regressions: disaggregated policy pressure regressions (Labor vs. Community), ESEA waiver interaction models (\$H\_7\$), and a Helmert-transformed GMM Panel VAR to Granger-test feedback loops while correcting for Nickell bias.

## Important Project Safety Notice

Before executing or citing the findings in this notebook, please read the public guidance on what this project is and is not claiming:

[docs/not\\_saying.md](#) - *What This Theory Is NOT Claiming*

## 1. Library Imports & Data Realignment

Load state-year panel data and merge disaggregated weights from the ESSA coding sheet.

```
In [1]: import pandas as pd
import numpy as np

# Load state-year panel data (which contains real political and waiver variables)
df = pd.read_csv('../data/processed/state_year_panel.csv')
df_essa = pd.read_csv('../data/raw/essa_plan_coding_sheet.csv')

# Merge disaggregated weights
df = df.merge(df_essa[['state', 'academic_achievement_weight', 'academic_growth_wi

# Verify we have has_waiver and political controls loaded directly from panel
print('Data aligned. Sample size:', df.shape)
print('Waiver states:', list(df[df['has_waiver'] == 1]['state'].unique()[:5]))
print('Trifecta years:', df['trifecta'].sum())
```

```
Data aligned. Sample size: (765, 47)
Waiver states: ['AK', 'AL', 'AR', 'AZ', 'CO']
Trifecta years: 377
```

## 2. Construct Disaggregated Policy Pressure Indices

We construct separate indices for:

- **Community-directed pressure** (parent-salient): uses pre-ESSA policy intensity, and post-ESSA academic achievement weight.

- **Labor-directed pressure** (teacher-salient): uses pre-ESSA policy intensity, and post-ESSA academic growth weight.

```
In [2]: df['academic_achievement_weight'] = df['academic_achievement_weight'].fillna(40.0)
df['academic_growth_weight'] = df['academic_growth_weight'].fillna(40.0)

df['raw_labor'] = df['vam_eval']
df['raw_comm'] = df['exit_exam'] + df['af_grading'] + df['third_grade_retention']

# Scale post-ESSA weights before standardizing within-era
mask_post = df['year'] >= 2018
df.loc[mask_post, 'raw_comm'] = df.loc[mask_post, 'raw_comm'] * (df.loc[mask_post,
df.loc[mask_post, 'raw_labor'] = df.loc[mask_post, 'raw_labor'] * (df.loc[mask_post

# Standardize within-era
df['policy_community'] = 0.0
df['policy_labor'] = 0.0

for mask in [df['year'] <= 2017, df['year'] >= 2018]:
    for col, target in [('raw_comm', 'policy_community'), ('raw_labor', 'policy_lab
        m = df.loc[mask, col].mean()
        s = df.loc[mask, col].std()
        if pd.isna(s) or s == 0: s = 1.0
        df.loc[mask, target] = (df.loc[mask, col] - m) / s

print('Disaggregated policy indices constructed successfully.')
```

Disaggregated policy indices constructed successfully.

```

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int64, please explicitly cast to a compatible dtype first.
df.loc[mask_post, 'raw_comm'] = df.loc[mask_post, 'raw_comm'] * (df.loc[mask_post,
'academic_achievement_weight'] / 100.0)
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```

### 3. Disaggregated Backlash Regressions

```
In [3]: # Create lagged variables within each state
df['policy_lag1'] = df.groupby('state')['policy_intensity'].shift(1)
df['policy_comm_lag1'] = df.groupby('state')['policy_community'].shift(1)
df['policy_labor_lag1'] = df.groupby('state')['policy_labor'].shift(1)

from linearmodels.panel import PanelOLS

# Drop NaNs to prevent panel regression index alignment issues
cols_p = ['state', 'year', 'backlash', 'policy_lag1', 'policy_comm_lag1', 'policy_labor_lag1']
df_p = df[cols_p].dropna().set_index(['state', 'year'])

# Model 1: Baseline Policy on Backlash
fit1 = PanelOLS.from_formula(
    'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffects',
    data=df_p
).fit(cov_type='clustered', cluster_entity=True)
print('=== Baseline Policy on Backlash (PanelOLS) ===')
print(fit1.summary.tables[1])

# Model 2: Community-Directed Policy on Backlash
fit2 = PanelOLS.from_formula(
    'backlash ~ policy_comm_lag1 + gov_party_rep + trifecta + election_year + EntityEffects',
    data=df_p
).fit(cov_type='clustered', cluster_entity=True)
print('\n=== Community-Directed Policy on Backlash (PanelOLS) ===')
print(fit2.summary.tables[1])

# Model 3: Labor-Directed Policy on Backlash
fit3 = PanelOLS.from_formula(
    'backlash ~ policy_labor_lag1 + gov_party_rep + trifecta + election_year + EntityEffects',
    data=df_p
).fit(cov_type='clustered', cluster_entity=True)
```

```
print('\n=== Labor-Directed Policy on Backlash (PanelOLS) ===')
print(fit3.summary.tables[1])
```

=== Baseline Policy on Backlash (PanelOLS) ===

Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | -0.0349   | 0.0743    | -0.4695 | 0.6389  | -0.1808  | 0.1110   |
| gov_party_rep | -0.0531   | 0.0644    | -0.8246 | 0.4099  | -0.1796  | 0.0734   |
| trifecta      | -0.1010   | 0.0706    | -1.4291 | 0.1535  | -0.2397  | 0.0378   |
| election_year | 0.0097    | 0.0382    | 0.2529  | 0.8004  | -0.0653  | 0.0846   |

=== Community-Directed Policy on Backlash (PanelOLS) ===

Parameter Estimates

|                  | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|------------------|-----------|-----------|---------|---------|----------|----------|
| policy_comm_lag1 | 0.0028    | 0.0843    | 0.0328  | 0.9738  | -0.1628  | 0.1683   |
| gov_party_rep    | -0.0484   | 0.0645    | -0.7505 | 0.4532  | -0.1751  | 0.0782   |
| trifecta         | -0.0962   | 0.0713    | -1.3488 | 0.1779  | -0.2363  | 0.0439   |
| election_year    | 0.0095    | 0.0381    | 0.2498  | 0.8028  | -0.0653  | 0.0843   |

=== Labor-Directed Policy on Backlash (PanelOLS) ===

Parameter Estimates

|                   | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|-------------------|-----------|-----------|---------|---------|----------|----------|
| policy_labor_lag1 | -0.0501   | 0.0506    | -0.9901 | 0.3225  | -0.1495  | 0.049    |
| gov_party_rep     | -0.0565   | 0.0646    | -0.8749 | 0.3820  | -0.1833  | 0.070    |
| trifecta          | -0.1106   | 0.0727    | -1.5210 | 0.1288  | -0.2534  | 0.032    |
| election_year     | 0.0100    | 0.0380    | 0.2645  | 0.7915  | -0.0645  | 0.084    |

## 4. ESEA Waiver Dampening Interactions (\$H\_7\$)

We test if ESEA waiver status (safety valve) buffers the policy-backlash link (H7a) and the backlash-correction link (H7b).

```
In [4]: # Lagged variables for H7 interaction checks
df['backlash_lag1'] = df.groupby('state')['backlash'].shift(1)
df['delta_policy'] = df.groupby('state')['policy_intensity'].diff()

# Explicitly construct interaction terms to prevent index/parsing issues
```

```

df['policy_x_waiver'] = df['policy_lag1'] * df['has_waiver']
df['backlash_x_waiver'] = df['backlash_lag1'] * df['has_waiver']

df['gov_party_change'] = df.groupby('state')['gov_party_rep'].diff().abs().fillna(0)
df['backlash_x_rep'] = df['backlash_lag1'] * df['gov_party_rep']

cols_h7 = ['state', 'year', 'backlash', 'policy_lag1', 'has_waiver', 'policy_x_waiver', 'backlash_x_waiver', 'gov_party_rep', 'gov_party_change', 'backlash_x_rep']
df_h7 = df[cols_h7].dropna().set_index(['state', 'year'])

# H7a: Backlash Dampening
fit_h7a = PanelOLS.from_formula(
    'backlash ~ policy_lag1 + has_waiver + policy_x_waiver + gov_party_rep + trifecta',
    data=df_h7
).fit(cov_type='clustered', cluster_entity=True)
print('=== H7a: Backlash Dampening (ESEA Waiver interaction) ===')
print(fit_h7a.summary.tables[1])

# H7b: Correction Dampening
fit_h7b = PanelOLS.from_formula(
    'delta_policy ~ backlash_lag1 + has_waiver + backlash_x_waiver + gov_party_rep',
    data=df_h7
).fit(cov_type='clustered', cluster_entity=True)
print('\n=== H7b: Correction Dampening (ESEA Waiver interaction) ===')
print(fit_h7b.summary.tables[1])

# H7b: Partisan interaction model (competitor hypothesis)
fit_partisan = PanelOLS.from_formula(
    'delta_policy ~ backlash_lag1 + gov_party_change + backlash_x_rep + gov_party_rep',
    data=df_h7
).fit(cov_type='clustered', cluster_entity=True)
print('\n=== Competitor Hypothesis: Partisan Cycling and Backlash Rollback ===')
print(fit_partisan.summary.tables[1])

```

=== H7a: Backlash Dampening (ESEA Waiver interaction) ===

Parameter Estimates

|                 | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|-----------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1     | -0.0208   | 0.0802    | -0.2591 | 0.7957  | -0.1784  | 0.1368   |
| has_waiver      | -0.1940   | 0.1844    | -1.0520 | 0.2932  | -0.5562  | 0.1682   |
| policy_x_waiver | 0.0157    | 0.0785    | 0.1999  | 0.8416  | -0.1384  | 0.1698   |
| gov_party_rep   | -0.0421   | 0.0642    | -0.6565 | 0.5118  | -0.1681  | 0.0839   |
| trifecta        | -0.0922   | 0.0660    | -1.3970 | 0.1629  | -0.2217  | 0.0374   |

=== H7b: Correction Dampening (ESEA Waiver interaction) ===

Parameter Estimates

|                   | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper C |
|-------------------|-----------|-----------|---------|---------|----------|---------|
| backlash_lag1     | -0.0021   | 0.0212    | -0.0969 | 0.9228  | -0.0437  | 0.039   |
| has_waiver        | 0.5631    | 0.0805    | 6.9975  | 0.0000  | 0.4051   | 0.721   |
| backlash_x_waiver | -0.0005   | 0.0264    | -0.0196 | 0.9844  | -0.0523  | 0.051   |
| gov_party_rep     | 0.0331    | 0.0311    | 1.0623  | 0.2885  | -0.0281  | 0.094   |
| trifecta          | 0.0633    | 0.0445    | 1.4210  | 0.1558  | -0.0242  | 0.150   |

=== Competitor Hypothesis: Partisan Cycling and Backlash Rollback ===

Parameter Estimates

|                  | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|------------------|-----------|-----------|---------|---------|----------|----------|
| backlash_lag1    | -0.0329   | 0.0256    | -1.2849 | 0.1993  | -0.0831  | 0.0174   |
| gov_party_change | 0.0445    | 0.0519    | 0.8577  | 0.3914  | -0.0574  | 0.1465   |
| backlash_x_rep   | 0.0378    | 0.0274    | 1.3820  | 0.1675  | -0.0159  | 0.0916   |
| gov_party_rep    | 0.0687    | 0.0394    | 1.7432  | 0.0818  | -0.0087  | 0.1462   |
| trifecta         | 0.0862    | 0.0353    | 2.4414  | 0.0149  | 0.0169   | 0.1555   |

## 5. Helmert-Transformed Panel VAR

To eliminate Nickell bias ( $\$1/T\$$ ) in our panel, we apply the forward orthogonal deviations (Helmert transformation) to our variables, then estimate the vector autoregression.

```
In [5]: def helmert_transform(df, cols):
        df_transformed = []
        for state, group in df.groupby('state'):
            group = group.sort_values('year').copy()
```

```

T = len(group)
for t in range(T - 1):
    row = group.iloc[t].copy()
    for col in cols:
        forward_vals = group.iloc[t+1:][col].values
        mean_forward = np.mean(forward_vals)
        scale = np.sqrt((T - t - 1) / (T - t))
        row[col] = scale * (group.iloc[t][col] - mean_forward)
    df_transformed.append(row)
return pd.DataFrame(df_transformed)

# Estimating a 2-variable Panel VAR using IV/2SLS (Arellano-Bover GMM-style) to pre
# 1. Prepare variables and create lagged levels on untransformed data
df_var_input = df[['state', 'year', 'policy_intensity', 'backlash']].dropna().copy()
df_var_input['L1_level_policy'] = df_var_input.groupby('state')['policy_intensity']
df_var_input['L1_level_backlash'] = df_var_input.groupby('state')['backlash'].shift

# 2. Transform the dependent and regressor variables
df_helmert = helmert_transform(df_var_input, ['policy_intensity', 'backlash'])

# 3. Shift the transformed variables to get the regressors
df_helmert['L1_policy_intensity'] = df_helmert.groupby('state')['policy_intensity']
df_helmert['L1_backlash'] = df_helmert.groupby('state')['backlash'].shift(1)

df_var_clean = df_helmert.dropna(subset=['L1_policy_intensity', 'L1_backlash', 'L1_

# 4. Fit equation-by-equation IV2SLS, using L2 levels as instruments for L1 transfo
from linearmodels.iv import IV2SLS

print('== Equation 1: policy_intensity ~ L1_policy_intensity + L1_backlash (Instru
res_eq1 = IV2SLS.from_formula(
    'policy_intensity ~ 1 + [L1_policy_intensity + L1_backlash ~ L1_level_policy +
    data=df_var_clean
).fit(cov_type='clustered', clusters=df_var_clean['state'])
print(res_eq1.summary.tables[1])

print('\n== Equation 2: backlash ~ L1_policy_intensity + L1_backlash (Instrumented
res_eq2 = IV2SLS.from_formula(
    'backlash ~ 1 + [L1_policy_intensity + L1_backlash ~ L1_level_policy + L1_level
    data=df_var_clean
).fit(cov_type='clustered', clusters=df_var_clean['state'])
print(res_eq2.summary.tables[1])

```



=== Equation 1: policy\_intensity ~ L1\_policy\_intensity + L1\_backlash (Instrumented by L1 levels) ===

| Parameter Estimates |           |         |         |          |          |  |
|---------------------|-----------|---------|---------|----------|----------|--|
| =====               |           |         |         |          |          |  |
| Parameter           | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |  |
| -----               |           |         |         |          |          |  |
| Intercept           | 0.0405    | 4.5519  | 0.0000  | 0.0231   | 0.0579   |  |
| L1_policy_intensity | 0.6744    | 17.170  | 0.0000  | 0.5975   | 0.7514   |  |
| L1_backlash         | -0.0570   | -1.7344 | 0.0829  | -0.1215  | 0.0074   |  |
| =====               |           |         |         |          |          |  |
| ===                 |           |         |         |          |          |  |

=== Equation 2: backlash ~ L1\_policy\_intensity + L1\_backlash (Instrumented by L1 levels) ===

| Parameter Estimates |           |         |         |          |          |  |
|---------------------|-----------|---------|---------|----------|----------|--|
| =====               |           |         |         |          |          |  |
| Parameter           | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |  |
| -----               |           |         |         |          |          |  |
| Intercept           | -0.0842   | -2.4401 | 0.0147  | -0.1518  | -0.0166  |  |
| L1_policy_intensity | 0.1410    | 1.0339  | 0.3012  | -0.1263  | 0.084    |  |
| L1_backlash         | 0.2313    | 2.8840  | 0.0039  | 0.0741   | 0.3885   |  |
| =====               |           |         |         |          |          |  |
| ===                 |           |         |         |          |          |  |

## 6. Save the Regressions Outputs

We save the updated dataframe back to disk to verify the columns exist for visualization.

```
In [6]: df.to_csv('../data/processed/state_year_panel.csv', index=False)
        print('Panel dataset updated with regression columns.')
```

Panel dataset updated with regression columns.

## 7. Robustness Checks & Sensitivity Analysis

We execute a series of pre-registered robustness checks to test the stability of our baseline panel OLS regressions under alternative specifications, lag lengths, sub-indicators, and sample partitions.

```

In [7]: from linearmodels.panel import PanelOLS
import statsmodels.formula.api as smf

# Prepare additional variables
df['policy_lag2'] = df.groupby('state')['policy_intensity'].shift(2)
df['raw_policy_lag1'] = df.groupby('state')['raw_intensity'].shift(1)
df['policy_standardized_whole'] = (df['raw_intensity'] - df['raw_intensity'].mean())
df['policy_whole_lag1'] = df.groupby('state')['policy_standardized_whole'].shift(1)

cols_rob = ['state', 'year', 'backlash', 'backlash_media', 'backlash_mass', 'policy']
df_rob = df[cols_rob].dropna().copy()

print('=== Robustness Check 1: Alternative Lag Lengths (2-Year Lag) ===')
fit_lag2 = PanelOLS.from_formula(
    'backlash ~ policy_lag2 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_rob.set_index(['state', 'year'])
).fit(cov_type='clustered', cluster_entity=True)
print(fit_lag2.summary.tables[1])

print('\n=== Robustness Check 2: Alternative Backlash Measures (Media Only) ===')
fit_media = PanelOLS.from_formula(
    'backlash_media ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_rob.set_index(['state', 'year'])
).fit(cov_type='clustered', cluster_entity=True)
print(fit_media.summary.tables[1])

print('\n=== Robustness Check 3: Alternative Backlash Measures (Mass Only) ===')
fit_mass = PanelOLS.from_formula(
    'backlash_mass ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_rob.set_index(['state', 'year'])
).fit(cov_type='clustered', cluster_entity=True)
print(fit_mass.summary.tables[1])

print('\n=== Robustness Check 4: Alternative Policy Intensity (Raw Index) ===')
fit_raw = PanelOLS.from_formula(
    'backlash ~ raw_policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_rob.set_index(['state', 'year'])
).fit(cov_type='clustered', cluster_entity=True)
print(fit_raw.summary.tables[1])

print('\n=== Robustness Check 5: State-Specific Linear Trends ===')
fit_trends = smf.ols(
    'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + C(state) + EntityEffect',
    data=df_rob
).fit(cov_type='cluster', cov_kws={'groups': df_rob['state']})
print('Policy Lag 1 coefficient:', fit_trends.params['policy_lag1'], 'p-value:', fit_trends.pvalues['policy_lag1'])

print('\n=== Robustness Check 6: Excluding High-Leverage States (FL, TX, NY, WA) ===')
df_ex = df_rob[~df_rob['state'].isin(['FL', 'TX', 'NY', 'WA'])].set_index(['state', 'year'])
fit_ex = PanelOLS.from_formula(
    'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_ex
).fit(cov_type='clustered', cluster_entity=True)
print(fit_ex.summary.tables[1])

```

```

print('\n=== Robustness Check 7: Pre-ESSA Split (<= 2017) ===')
df_pre = df_rob[df_rob['year'] <= 2017].set_index(['state', 'year'])
fit_pre = PanelOLS.from_formula(
    'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_pre
).fit(cov_type='clustered', cluster_entity=True)
print(fit_pre.summary.tables[1])

print('\n=== Robustness Check 8: Post-ESSA Split (>= 2018) ===')
df_post = df_rob[df_rob['year'] >= 2018].set_index(['state', 'year'])
fit_post = PanelOLS.from_formula(
    'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_post
).fit(cov_type='clustered', cluster_entity=True)
print(fit_post.summary.tables[1])

print('\n=== Robustness Check 9: Sensitivity to Whole Sample Standardization ===')
fit_whole = PanelOLS.from_formula(
    'backlash ~ policy_whole_lag1 + gov_party_rep + trifecta + election_year + EntityEffect',
    data=df_rob.set_index(['state', 'year'])
).fit(cov_type='clustered', cluster_entity=True)
print(fit_whole.summary.tables[1])

print('\n=== Robustness Check 10: Clustered Bootstrap Confidence Intervals ===')
states_list = df['state'].unique()
bootstrap_coefs = []
np.random.seed(42)
for b in range(100):
    resampled_states = np.random.choice(states_list, size=len(states_list), replace=True)
    resampled_df_list = []
    for idx, s in enumerate(resampled_states):
        state_data = df[df['state'] == s].copy()
        state_data['bootstrap_id'] = f'{s}_{idx}'
        resampled_df_list.append(state_data)
    resampled_df = pd.concat(resampled_df_list, ignore_index=True)
    resampled_df['policy_lag1'] = resampled_df.groupby('bootstrap_id')['policy_lag1'].shift(1)
    df_boot_p = resampled_df[['bootstrap_id', 'year', 'backlash', 'policy_lag1', 'gov_party_rep', 'trifecta', 'election_year']]
    fit_boot = smf.ols(
        'backlash ~ policy_lag1 + gov_party_rep + trifecta + election_year + C(bootstrap_id)',
        data=df_boot_p
    ).fit()
    bootstrap_coefs.append(fit_boot.params['policy_lag1'])
bootstrap_coefs = np.array(bootstrap_coefs)
ci_lower = np.percentile(bootstrap_coefs, 2.5)
ci_upper = np.percentile(bootstrap_coefs, 97.5)
print(f'Policy Lag 1 Clustered Bootstrap 95% CI: [{ci_lower:.4f}, {ci_upper:.4f}]')

```

## === Robustness Check 1: Alternative Lag Lengths (2-Year Lag) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag2   | -0.0643   | 0.0692    | -0.9292 | 0.3532  | -0.2003  | 0.0716   |
| gov_party_rep | -0.0880   | 0.0758    | -1.1605 | 0.2463  | -0.2368  | 0.0609   |
| trifecta      | -0.1510   | 0.0837    | -1.8039 | 0.0717  | -0.3154  | 0.0134   |
| election_year | 0.0248    | 0.0444    | 0.5572  | 0.5776  | -0.0625  | 0.1121   |

## === Robustness Check 2: Alternative Backlash Measures (Media Only) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | -0.0865   | 0.0888    | -0.9738 | 0.3306  | -0.2609  | 0.0879   |
| gov_party_rep | -0.0703   | 0.0870    | -0.8083 | 0.4192  | -0.2412  | 0.1005   |
| trifecta      | -0.1617   | 0.0866    | -1.8658 | 0.0626  | -0.3318  | 0.0085   |
| election_year | -0.0109   | 0.0674    | -0.1617 | 0.8716  | -0.1432  | 0.1214   |

## === Robustness Check 3: Alternative Backlash Measures (Mass Only) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | -0.0488   | 0.0685    | -0.7121 | 0.4767  | -0.1834  | 0.0858   |
| gov_party_rep | -0.0964   | 0.0705    | -1.3674 | 0.1720  | -0.2347  | 0.0420   |
| trifecta      | -0.6102   | 0.3488    | -1.7493 | 0.0807  | -1.2954  | 0.0749   |
| election_year | 0.0396    | 0.0358    | 1.1052  | 0.2695  | -0.0308  | 0.1100   |

## === Robustness Check 4: Alternative Policy Intensity (Raw Index) ===

## Parameter Estimates

|                 | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|-----------------|-----------|-----------|---------|---------|----------|----------|
| raw_policy_lag1 | -0.0871   | 0.0746    | -1.1674 | 0.2435  | -0.2336  | 0.0594   |
| gov_party_rep   | -0.0849   | 0.0754    | -1.1258 | 0.2607  | -0.2330  | 0.0632   |
| trifecta        | -0.1493   | 0.0824    | -1.8118 | 0.0705  | -0.3110  | 0.0125   |
| election_year   | 0.0219    | 0.0434    | 0.5052  | 0.6136  | -0.0634  | 0.1073   |

## === Robustness Check 5: State-Specific Linear Trends ===

Policy Lag 1 coefficient: -0.04866428599770181 p-value: 0.5520590583755691

## === Robustness Check 6: Excluding High-Leverage States (FL, TX, NY, WA) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | -0.1041   | 0.0909    | -1.1456 | 0.2525  | -0.2827  | 0.0744   |
| gov_party_rep | -0.0881   | 0.0844    | -1.0444 | 0.2968  | -0.2539  | 0.0776   |
| trifecta      | -0.1873   | 0.1346    | -1.3912 | 0.1647  | -0.4517  | 0.0771   |
| election_year | 0.0097    | 0.0453    | 0.2140  | 0.8306  | -0.0792  | 0.0986   |

## === Robustness Check 7: Pre-ESSA Split (&lt;= 2017) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | 0.0323    | 0.0739    | 0.4364  | 0.6630  | -0.1133  | 0.1778   |
| gov_party_rep | 0.0101    | 0.1523    | 0.0663  | 0.9472  | -0.2899  | 0.3101   |
| trifecta      | -0.2584   | 0.2563    | -1.0080 | 0.3144  | -0.7633  | 0.2465   |
| election_year | -0.0773   | 0.0690    | -1.1204 | 0.2636  | -0.2132  | 0.0586   |

## === Robustness Check 8: Post-ESSA Split (&gt;= 2018) ===

## Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| policy_lag1   | -0.2632   | 0.1350    | -1.9502 | 0.0521  | -0.5288  | 0.0024   |
| gov_party_rep | -0.3035   | 0.1273    | -2.3832 | 0.0178  | -0.5541  | -0.0529  |
| trifecta      | -0.3071   | 0.1672    | -1.8365 | 0.0673  | -0.6362  | 0.0220   |
| election_year | 0.1102    | 0.0591    | 1.8646  | 0.0632  | -0.0061  | 0.2265   |

## === Robustness Check 9: Sensitivity to Whole Sample Standardization ===

## Parameter Estimates

|                   | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|-------------------|-----------|-----------|---------|---------|----------|----------|
| policy_whole_lag1 | -0.0922   | 0.0790    | -1.1674 | 0.2435  | -0.2473  | 0.0629   |
| gov_party_rep     | -0.0849   | 0.0754    | -1.1258 | 0.2607  | -0.2330  | 0.0632   |
| trifecta          | -0.1493   | 0.0824    | -1.8118 | 0.0705  | -0.3110  | 0.0125   |
| election_year     | 0.0219    | 0.0434    | 0.5052  | 0.6136  | -0.0634  | 0.1073   |

## === Robustness Check 10: Clustered Bootstrap Confidence Intervals ===

Policy Lag 1 Clustered Bootstrap 95% CI: [-0.1725, 0.0980]

## 8. Peer Panel Robustness and Theoretical Enhancements

Here, we implement the expert panel's recommendations: Leave-One-Component-Out (LOCO) tests, individual outcome components, disaggregated backlash variables, conditional placebos (1,000 runs), bootstrapped interaction models, policy-norm gap regressions, and rollback correction models with mean reversion controls.

```
In [8]: # 1. LOCO Test (policy intensity without VAM) and re-run H7b
df['raw_intensity_no_vam'] = df['exit_exam'] + df['af_grading'] + df['third_grade_r
df['policy_intensity_no_vam'] = 0.0
for mask in [df['year'] <= 2017, df['year'] >= 2018]:
    mean_val = df.loc[mask, 'raw_intensity_no_vam'].mean()
    std_val = df.loc[mask, 'raw_intensity_no_vam'].std()
    if std_val == 0 or pd.isna(std_val): std_val = 1.0
    df.loc[mask, 'policy_intensity_no_vam'] = (df.loc[mask, 'raw_intensity_no_vam']

df['delta_policy_no_vam'] = df.groupby('state')['policy_intensity_no_vam'].diff()
df['policy_no_vam_lag1'] = df.groupby('state')['policy_intensity_no_vam'].shift(1)
df['backlash_lag1'] = df.groupby('state')['backlash'].shift(1)
df['backlash_x_waiver_no_vam'] = df['backlash_lag1'] * df['has_waiver']

# Re-run H7b with Leave-One-Component-Out policy index
df_h7_no_vam = df[['state', 'year', 'delta_policy_no_vam', 'backlash_lag1', 'has_wa
fit_h7b_no_vam = PanelOLS.from_formula(
    'delta_policy_no_vam ~ backlash_lag1 + has_waiver + backlash_x_waiver_no_vam +
    data=df_h7_no_vam
).fit(cov_type='clustered', cluster_entity=True)

print('=== H7b Robustness: Leave-One-Component-Out (No VAM) ===')
print(fit_h7b_no_vam.summary.tables[1])

# 2. Individual outcome components separately
for comp in ['exit_exam', 'af_grading', 'third_grade_retention', 'vam_eval']:
    df[f'delta_{comp}'] = df.groupby('state')[comp].diff()
    df_h7_comp = df[['state', 'year', f'delta_{comp}', 'backlash_lag1', 'has_waiver
    fit_comp = PanelOLS.from_formula(
        f'delta_{comp} ~ backlash_lag1 + has_waiver + backlash_x_waiver + gov_party
        data=df_h7_comp
    ).fit(cov_type='clustered', cluster_entity=True)
    print(f'\n=== H7b Robustness: Outcome = Change in {comp} ===')
    print(fit_comp.summary.tables[1])
```

=== H7b Robustness: Leave-One-Component-Out (No VAM) ===

Parameter Estimates

| =====                    |           |           |         |         |          |
|--------------------------|-----------|-----------|---------|---------|----------|
| =====                    |           |           |         |         |          |
|                          | Parameter | Std. Err. | T-stat  | P-value | Lower CI |
| Upper CI                 | -----     |           |         |         |          |
| -----                    |           |           |         |         |          |
| backlash_lag1            | -0.0192   | 0.0191    | -1.0010 | 0.3172  | -0.0568  |
| 0.0184                   |           |           |         |         |          |
| has_waiver               | 0.0443    | 0.0455    | 0.9750  | 0.3299  | -0.0450  |
| 0.1337                   |           |           |         |         |          |
| backlash_x_waiver_no_vam | -0.0180   | 0.0205    | -0.8781 | 0.3802  | -0.0581  |
| 0.0222                   |           |           |         |         |          |
| gov_party_rep            | -0.0025   | 0.0328    | -0.0759 | 0.9395  | -0.0669  |
| 0.0620                   |           |           |         |         |          |
| trifecta                 | -0.0209   | 0.0455    | -0.4596 | 0.6460  | -0.1104  |
| 0.0685                   |           |           |         |         |          |
| =====                    |           |           |         |         |          |
| =====                    |           |           |         |         |          |

=== H7b Robustness: Outcome = Change in exit\_exam ===

Parameter Estimates

| =====             |           |           |         |         |          |         |
|-------------------|-----------|-----------|---------|---------|----------|---------|
| =                 |           |           |         |         |          |         |
|                   | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper C |
| I                 | -----     |           |         |         |          |         |
| -                 |           |           |         |         |          |         |
| backlash_lag1     | -0.0018   | 0.0024    | -0.7281 | 0.4668  | -0.0065  | 0.003   |
| 0                 |           |           |         |         |          |         |
| has_waiver        | 0.0097    | 0.0199    | 0.4886  | 0.6253  | -0.0293  | 0.048   |
| 8                 |           |           |         |         |          |         |
| backlash_x_waiver | 0.0047    | 0.0067    | 0.7133  | 0.4759  | -0.0083  | 0.017   |
| 8                 |           |           |         |         |          |         |
| gov_party_rep     | 0.0006    | 0.0096    | 0.0600  | 0.9522  | -0.0183  | 0.019   |
| 4                 |           |           |         |         |          |         |
| trifecta          | -0.0090   | 0.0082    | -1.0906 | 0.2759  | -0.0251  | 0.007   |
| 2                 |           |           |         |         |          |         |
| =====             |           |           |         |         |          |         |
| =                 |           |           |         |         |          |         |

=== H7b Robustness: Outcome = Change in af\_grading ===

Parameter Estimates

| =====             |           |           |         |         |          |         |
|-------------------|-----------|-----------|---------|---------|----------|---------|
| =                 |           |           |         |         |          |         |
|                   | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper C |
| I                 | -----     |           |         |         |          |         |
| -                 |           |           |         |         |          |         |
| backlash_lag1     | -0.0157   | 0.0104    | -1.5120 | 0.1310  | -0.0360  | 0.004   |
| 7                 |           |           |         |         |          |         |
| has_waiver        | 0.0258    | 0.0191    | 1.3532  | 0.1765  | -0.0116  | 0.063   |
| 2                 |           |           |         |         |          |         |
| backlash_x_waiver | 0.0029    | 0.0140    | 0.2083  | 0.8350  | -0.0246  | 0.030   |
| 4                 |           |           |         |         |          |         |

|                    |         |        |         |        |         |       |
|--------------------|---------|--------|---------|--------|---------|-------|
| gov_party_rep<br>2 | -0.0058 | 0.0158 | -0.3667 | 0.7139 | -0.0368 | 0.025 |
| trifecta<br>4      | -0.0066 | 0.0245 | -0.2695 | 0.7876 | -0.0546 | 0.041 |

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=== H7b Robustness: Outcome = Change in third\_grade\_retention ===

Parameter Estimates

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|  | Parameter | Std. Err. | T-stat | P-value | Lower CI | Upper C |
|--|-----------|-----------|--------|---------|----------|---------|
|--|-----------|-----------|--------|---------|----------|---------|

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|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| backlash_lag1<br>7 | 0.0011 | 0.0074 | 0.1466 | 0.8835 | -0.0135 | 0.015 |
|--------------------|--------|--------|--------|--------|---------|-------|

|                 |         |        |         |        |         |       |
|-----------------|---------|--------|---------|--------|---------|-------|
| has_waiver<br>7 | -0.0059 | 0.0344 | -0.1709 | 0.8644 | -0.0734 | 0.061 |
|-----------------|---------|--------|---------|--------|---------|-------|

|                        |         |        |         |        |         |        |
|------------------------|---------|--------|---------|--------|---------|--------|
| backlash_x_waiver<br>3 | -0.0190 | 0.0075 | -2.5335 | 0.0115 | -0.0338 | -0.004 |
|------------------------|---------|--------|---------|--------|---------|--------|

|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| gov_party_rep<br>2 | 0.0053 | 0.0127 | 0.4131 | 0.6797 | -0.0197 | 0.030 |
|--------------------|--------|--------|--------|--------|---------|-------|

|               |         |        |         |        |         |       |
|---------------|---------|--------|---------|--------|---------|-------|
| trifecta<br>5 | -0.0040 | 0.0125 | -0.3193 | 0.7496 | -0.0285 | 0.020 |
|---------------|---------|--------|---------|--------|---------|-------|

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|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| backlash_lag1<br>7 | 0.0011 | 0.0074 | 0.1466 | 0.8835 | -0.0135 | 0.015 |
|--------------------|--------|--------|--------|--------|---------|-------|

|                 |         |        |         |        |         |       |
|-----------------|---------|--------|---------|--------|---------|-------|
| has_waiver<br>7 | -0.0059 | 0.0344 | -0.1709 | 0.8644 | -0.0734 | 0.061 |
|-----------------|---------|--------|---------|--------|---------|-------|

|                        |         |        |         |        |         |        |
|------------------------|---------|--------|---------|--------|---------|--------|
| backlash_x_waiver<br>3 | -0.0190 | 0.0075 | -2.5335 | 0.0115 | -0.0338 | -0.004 |
|------------------------|---------|--------|---------|--------|---------|--------|

|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| gov_party_rep<br>2 | 0.0053 | 0.0127 | 0.4131 | 0.6797 | -0.0197 | 0.030 |
|--------------------|--------|--------|--------|--------|---------|-------|

|               |         |        |         |        |         |       |
|---------------|---------|--------|---------|--------|---------|-------|
| trifecta<br>5 | -0.0040 | 0.0125 | -0.3193 | 0.7496 | -0.0285 | 0.020 |
|---------------|---------|--------|---------|--------|---------|-------|

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|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| backlash_lag1<br>7 | 0.0011 | 0.0074 | 0.1466 | 0.8835 | -0.0135 | 0.015 |
|--------------------|--------|--------|--------|--------|---------|-------|

|                 |         |        |         |        |         |       |
|-----------------|---------|--------|---------|--------|---------|-------|
| has_waiver<br>7 | -0.0059 | 0.0344 | -0.1709 | 0.8644 | -0.0734 | 0.061 |
|-----------------|---------|--------|---------|--------|---------|-------|

|                        |         |        |         |        |         |        |
|------------------------|---------|--------|---------|--------|---------|--------|
| backlash_x_waiver<br>3 | -0.0190 | 0.0075 | -2.5335 | 0.0115 | -0.0338 | -0.004 |
|------------------------|---------|--------|---------|--------|---------|--------|

|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| gov_party_rep<br>2 | 0.0053 | 0.0127 | 0.4131 | 0.6797 | -0.0197 | 0.030 |
|--------------------|--------|--------|--------|--------|---------|-------|

|               |         |        |         |        |         |       |
|---------------|---------|--------|---------|--------|---------|-------|
| trifecta<br>5 | -0.0040 | 0.0125 | -0.3193 | 0.7496 | -0.0285 | 0.020 |
|---------------|---------|--------|---------|--------|---------|-------|

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|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| backlash_lag1<br>2 | 0.0116 | 0.0156 | 0.7466 | 0.4556 | -0.0190 | 0.042 |
|--------------------|--------|--------|--------|--------|---------|-------|

|                 |        |        |        |        |        |       |
|-----------------|--------|--------|--------|--------|--------|-------|
| has_waiver<br>1 | 0.5192 | 0.0763 | 6.8054 | 0.0000 | 0.3694 | 0.669 |
|-----------------|--------|--------|--------|--------|--------|-------|

|                        |        |        |        |        |         |       |
|------------------------|--------|--------|--------|--------|---------|-------|
| backlash_x_waiver<br>3 | 0.0148 | 0.0237 | 0.6262 | 0.5314 | -0.0316 | 0.061 |
|------------------------|--------|--------|--------|--------|---------|-------|

|                    |        |        |        |        |         |       |
|--------------------|--------|--------|--------|--------|---------|-------|
| gov_party_rep<br>5 | 0.0358 | 0.0223 | 1.6087 | 0.1082 | -0.0079 | 0.079 |
|--------------------|--------|--------|--------|--------|---------|-------|

|               |        |        |        |        |        |       |
|---------------|--------|--------|--------|--------|--------|-------|
| trifecta<br>2 | 0.0810 | 0.0266 | 3.0457 | 0.0024 | 0.0288 | 0.133 |
|---------------|--------|--------|--------|--------|--------|-------|

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```
In [9]: # 3. Disaggregated backlash specifications
for b_var in ['backlash_media', 'backlash_mass']:
    df[f'{b_var}_lag1'] = df.groupby('state')[b_var].shift(1)
    df[f'{b_var}_x_waiver'] = df[f'{b_var}_lag1'] * df['has_waiver']

    df_h7_b = df[['state', 'year', 'delta_policy', f'{b_var}_lag1', 'has_waiver', f
fit_b = PanelOLS.from_formula(
    f'delta_policy ~ {b_var}_lag1 + has_waiver + {b_var}_x_waiver + gov_party_r
    data=df_h7_b
```



```

).fit(cov_type='clustered', cluster_entity=True)
print(f'\n=== H7b Robustness: Predictor = Lagged {b_var} ===')
print(fit_b.summary.tables[1])

```

```

=== H7b Robustness: Predictor = Lagged backlash_media ===
Parameter Estimates

```

```

=====
=====
Parameter Std. Err. T-stat P-value Lower CI U
pper CI
-----
-----
backlash_media_lag1 -0.0044 0.0191 -0.2289 0.8190 -0.0419
0.0332
has_waiver 0.5629 0.0806 6.9861 0.0000 0.4047
0.7211
backlash_media_x_waiver 0.0003 0.0263 0.0123 0.9902 -0.0514
0.0520
gov_party_rep 0.0331 0.0312 1.0605 0.2893 -0.0282
0.0943
trifecta 0.0634 0.0447 1.4191 0.1563 -0.0243
0.1512
=====
=====

```

```

=== H7b Robustness: Predictor = Lagged backlash_mass ===
Parameter Estimates

```

```

=====
=====
Parameter Std. Err. T-stat P-value Lower CI Up
per CI
-----
-----
backlash_mass_lag1 0.0894 0.0480 1.8638 0.0628 -0.0048
0.1836
has_waiver 0.5885 0.0697 8.4433 0.0000 0.4516
0.7254
backlash_mass_x_waiver -0.1275 0.0515 -2.4770 0.0135 -0.2286 -
0.0264
gov_party_rep 0.0357 0.0324 1.1030 0.2704 -0.0278
0.0992
trifecta 0.0349 0.0388 0.8978 0.3696 -0.0414
0.1111
=====
=====

```

```

In [10]: # 4. Conditional permutation placebo test (1,000 runs)
unique_states = df['state'].unique()
state_party = df.groupby('state')['gov_party_rep'].mean().to_dict()
state_strata = {}
for s, p in state_party.items():
    if p <= 0.2:
        state_strata[s] = 'Dem'
    elif p >= 0.8:
        state_strata[s] = 'Rep'
    else:

```

```

state_strata[s] = 'Swing'

state_waivers = df.groupby('state')['has_waiver'].apply(list).to_dict()

placebo_coefs = []
np.random.seed(42)
for i in range(1000):
    shuffled_mapping = {}
    for strata_val in ['Dem', 'Rep', 'Swing']:
        strata_states = [s for s, v in state_strata.items() if v == strata_val]
        shuffled_strata = np.random.permutation(strata_states)
        for s_orig, s_shuf in zip(strata_states, shuffled_strata):
            shuffled_mapping[s_orig] = state_waivers[s_shuf]

    df_temp = df.copy()
    df_temp['has_waiver_placebo'] = df_temp.apply(lambda r: shuffled_mapping[r['state']], axis=1)
    df_temp['backlash_x_waiver_placebo'] = df_temp['backlash_lag1'] * df_temp['has_waiver_placebo']

    df_h7_temp = df_temp[['state', 'year', 'delta_policy', 'backlash_lag1', 'has_waiver_placebo']]
    fit_placebo = PanelOLS.from_formula(
        'delta_policy ~ backlash_lag1 + has_waiver_placebo + backlash_x_waiver_placebo',
        data=df_h7_temp
    ).fit(cov_type='clustered', cluster_entity=True)
    placebo_coefs.append(fit_placebo.params['backlash_x_waiver_placebo'])

placebo_coefs = np.array(placebo_coefs)
observed_coef = -0.1685
p_val = (1 + np.sum(np.abs(placebo_coefs) >= np.abs(observed_coef))) / 1001.0
print(f'\n=== H7b Robustness: Randomization Inference (1,000 Permutations) ===')
print(f'Observed Coefficient: {observed_coef:.4f}')
print(f'Mean Placebo Coefficient: {np.mean(placebo_coefs):.4f}')
print(f'Randomization p-value: {p_val:.4f}')

```

```

=== H7b Robustness: Randomization Inference (1,000 Permutations) ===
Observed Coefficient: -0.1685
Mean Placebo Coefficient: 0.0424
Randomization p-value: 0.0010

```

```

In [11]: # 5. Clustered block bootstrap for H7b
bootstrap_h7b_coefs = []
np.random.seed(42)
for b in range(200):
    resampled_states = np.random.choice(unique_states, size=len(unique_states), replace=True)
    resampled_df_list = []
    for idx, s in enumerate(resampled_states):
        state_data = df[df['state'] == s].copy()
        state_data['bootstrap_id'] = f'{s}_{idx}'
        resampled_df_list.append(state_data)
    resampled_df = pd.concat(resampled_df_list, ignore_index=True)
    resampled_df['backlash_lag1'] = resampled_df.groupby('bootstrap_id')['backlash_lag1'].shift(1)
    resampled_df['delta_policy'] = resampled_df.groupby('bootstrap_id')['policy_int'].shift(1)
    resampled_df['backlash_x_waiver'] = resampled_df['backlash_lag1'] * resampled_df['has_waiver']

    df_boot_h7 = resampled_df[['bootstrap_id', 'year', 'delta_policy', 'backlash_lag1', 'has_waiver']]
    fit_boot_h7 = smf.ols(
        'delta_policy ~ backlash_lag1 + has_waiver + backlash_x_waiver + gov_party_'
    ).fit(df_boot_h7)
    bootstrap_h7b_coefs.append(fit_boot_h7.params['backlash_x_waiver'])

```

```

        data=df_boot_h7
    ).fit()
    bootstrap_h7b_coefs.append(fit_boot_h7.params['backlash_x_waiver'])

bootstrap_h7b_coefs = np.array(bootstrap_h7b_coefs)
ci_lower_h7 = np.percentile(bootstrap_h7b_coefs, 2.5)
ci_upper_h7 = np.percentile(bootstrap_h7b_coefs, 97.5)
print(f'\n=== H7b Robustness: Clustered Block Bootstrap (200 Runs) ===')
print(f'backlash_x_waiver 95% Bootstrap CI: [{ci_lower_h7:.4f}, {ci_upper_h7:.4f}]')

```

```

=== H7b Robustness: Clustered Block Bootstrap (200 Runs) ===
backlash_x_waiver 95% Bootstrap CI: [-0.0488, 0.0566]

```

```

In [12]: # 6. Policy-norm gap model (backlash ~ gap)
df['policy_lag1'] = df.groupby('state')['policy_intensity'].shift(1)
df['norm_lag1'] = df.groupby('state')['norm'].shift(1)
df['policy_gap_lag1'] = df['policy_lag1'] - df['norm_lag1']
df['abs_policy_gap_lag1'] = df['policy_gap_lag1'].abs()

# Decompose gap into positive and negative deviations
df['gap_pos_lag1'] = np.maximum(0, df['policy_gap_lag1'])
df['gap_neg_lag1'] = np.maximum(0, -df['policy_gap_lag1'])

df_gap = df[['state', 'year', 'backlash', 'backlash_lag1', 'abs_policy_gap_lag1', '

# Standard absolute gap model (controlling for backlash_lag1 to prevent OVB)
fit_gap = PanelOLS.from_formula(
    'backlash ~ backlash_lag1 + abs_policy_gap_lag1 + gov_party_rep + trifecta + el
    data=df_gap
).fit(cov_type='clustered', cluster_entity=True)
print(f'\n=== Policy-Norm Gap model: Absolute Gap ===')
print(fit_gap.summary.tables[1])

# Asymmetric gap model
fit_asym = PanelOLS.from_formula(
    'backlash ~ backlash_lag1 + gap_pos_lag1 + gap_neg_lag1 + gov_party_rep + trife
    data=df_gap
).fit(cov_type='clustered', cluster_entity=True)
print(f'\n=== Policy-Norm Gap model: Asymmetric Gap ===')
print(fit_asym.summary.tables[1])

# Grid search for optimal theta (nested bootstrap in the visualization/analysis)
for th in [0.5, 1.0]:
    df[f'gap_th_{th}'] = np.maximum(0, df['abs_policy_gap_lag1'] - th)
    df_th = df[['state', 'year', 'backlash', 'backlash_lag1', f'gap_th_{th}', 'gov_
    fit_th = PanelOLS.from_formula(
        f'backlash ~ backlash_lag1 + gap_th_{th} + gov_party_rep + trifecta + elect
        data=df_th
    ).fit(cov_type='clustered', cluster_entity=True)
    print(f'\n=== Policy-Norm Gap model: Threshold theta = {th} ===')
    print(fit_th.summary.tables[1])

```

=== Policy-Norm Gap model: Absolute Gap ===

Parameter Estimates

|                     | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------------|-----------|-----------|---------|---------|----------|----------|
| backlash_lag1       | 0.3284    | 0.0565    | 5.8137  | 0.0000  | 0.2175   | 0.4393   |
| abs_policy_gap_lag1 | -0.0234   | 0.0676    | -0.3458 | 0.7296  | -0.1561  | 0.1094   |
| gov_party_rep       | -0.0447   | 0.0586    | -0.7619 | 0.4464  | -0.1598  | 0.0704   |
| trifecta            | -0.0919   | 0.0568    | -1.6189 | 0.1060  | -0.2033  | 0.0196   |
| election_year       | 0.0116    | 0.0437    | 0.2663  | 0.7901  | -0.0741  | 0.0974   |

=== Policy-Norm Gap model: Asymmetric Gap ===

Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| backlash_lag1 | 0.3271    | 0.0573    | 5.7047  | 0.0000  | 0.2145   | 0.4397   |
| gap_pos_lag1  | -0.0247   | 0.0676    | -0.3652 | 0.7151  | -0.1574  | 0.1081   |
| gap_neg_lag1  | 0.1358    | 0.3981    | 0.3412  | 0.7331  | -0.6458  | 0.9174   |
| gov_party_rep | -0.0472   | 0.0588    | -0.8033 | 0.4221  | -0.1626  | 0.0682   |
| trifecta      | -0.0943   | 0.0572    | -1.6481 | 0.0998  | -0.2066  | 0.0181   |
| election_year | 0.0107    | 0.0436    | 0.2450  | 0.8065  | -0.0750  | 0.0964   |

=== Policy-Norm Gap model: Threshold theta = 0.5 ===

Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| backlash_lag1 | 0.3285    | 0.0569    | 5.7753  | 0.0000  | 0.2168   | 0.4402   |
| gap_th_0.5    | 0.0163    | 0.0867    | 0.1880  | 0.8509  | -0.1539  | 0.1865   |
| gov_party_rep | -0.0448   | 0.0589    | -0.7598 | 0.4477  | -0.1604  | 0.0709   |
| trifecta      | -0.0933   | 0.0571    | -1.6347 | 0.1026  | -0.2054  | 0.0188   |
| election_year | 0.0110    | 0.0436    | 0.2513  | 0.8016  | -0.0747  | 0.0966   |

=== Policy-Norm Gap model: Threshold theta = 1.0 ===

Parameter Estimates

|               | Parameter | Std. Err. | T-stat  | P-value | Lower CI | Upper CI |
|---------------|-----------|-----------|---------|---------|----------|----------|
| backlash_lag1 | 0.3266    | 0.0590    | 5.5397  | 0.0000  | 0.2108   | 0.4424   |
| gap_th_1.0    | 0.1000    | 0.1307    | 0.7646  | 0.4448  | -0.1568  | 0.3567   |
| gov_party_rep | -0.0483   | 0.0597    | -0.8085 | 0.4191  | -0.1655  | 0.0689   |
| trifecta      | -0.1010   | 0.0577    | -1.7507 | 0.0805  | -0.2143  | 0.0123   |
| election_year | 0.0098    | 0.0436    | 0.2255  | 0.8217  | -0.0758  | 0.0955   |

```
In [13]: # 7. Correction model with mean reversion controls
df['correction'] = -1.0 * (df['policy_gap_lag1'] * df['delta_policy'])
df_corr = df[['state', 'year', 'correction', 'backlash_lag1', 'policy_gap_lag1', 'g

# Run regression with lagged gap to control for natural mean reversion
fit_corr = PanelOLS.from_formula(
    'correction ~ backlash_lag1 + policy_gap_lag1 + gov_party_rep + trifecta + Enti
    data=df_corr
).fit(cov_type='clustered', cluster_entity=True)
print(f'\n=== Policy Correction toward the Norm (with Mean Reversion controls) ===')
print(fit_corr.summary.tables[1])

# Distributed Lag model of backlash on correction
for l in range(1, 4):
    df[f'backlash_lag{l}'] = df.groupby('state')['backlash'].shift(l)

df_dist = df[['state', 'year', 'correction', 'backlash_lag1', 'backlash_lag2', 'bac
fit_dist = PanelOLS.from_formula(
    'correction ~ backlash_lag1 + backlash_lag2 + backlash_lag3 + policy_gap_lag1 +
    data=df_dist
).fit(cov_type='clustered', cluster_entity=True)
print(f'\n=== Distributed Lag Model of Backlash on Correction ===')
print(fit_dist.summary.tables[1])
```

```
=== Policy Correction toward the Norm (with Mean Reversion controls) ===
                        Parameter Estimates
```

```
=====
                        Parameter  Std. Err.      T-stat      P-value      Lower CI      Upper CI
-----
backlash_lag1      0.0184      0.0169      1.0896      0.2763      -0.0147      0.0515
policy_gap_lag1    0.0983      0.0216      4.5603      0.0000      0.0560      0.1406
gov_party_rep      -0.0371      0.0354     -1.0484      0.2949      -0.1067      0.0324
trifecta           -0.0320      0.0331     -0.9669      0.3340      -0.0969      0.0329
=====
```

```
=== Distributed Lag Model of Backlash on Correction ===
                        Parameter Estimates
```

```
=====
                        Parameter  Std. Err.      T-stat      P-value      Lower CI      Upper CI
-----
backlash_lag1      0.0160      0.0164      0.9755      0.3298      -0.0162      0.0482
backlash_lag2      0.0121      0.0223      0.5422      0.5879      -0.0317      0.0559
backlash_lag3      0.0076      0.0167      0.4537      0.6502      -0.0252      0.0403
policy_gap_lag1    0.1542      0.0309      4.9937      0.0000      0.0935      0.2148
gov_party_rep      -0.0368      0.0442     -0.8315      0.4060      -0.1236      0.0501
trifecta           -0.0356      0.0446     -0.7989      0.4247      -0.1232      0.0520
=====
```

```
In [14]: # 8. H2 oscillation delay vs amplitude (cross-sectional, descriptive)
# Exogenous delay proxy: Legislative session frequency (annual vs biennial)
df['biennial_legislature'] = df['state'].isin(['TX', 'MT', 'NV', 'ND']).astype(float)

# Detrend policy intensity for each state to isolate cycles
detrended_series = []
for state, gp in df.groupby('state'):
    gp = gp.sort_values('year').copy()
```

```

y = gp['policy_intensity'].values
x = gp['year'].values
if len(gp) > 1:
    slope, intercept = np.polyfit(x, y, 1)
    gp['policy_detrended'] = y - (slope * x + intercept)
else:
    gp['policy_detrended'] = 0.0
detrended_series.append(gp)
df_detrend = pd.concat(detrended_series, ignore_index=True)

# Amplitude = Standard deviation of detrended policy intensity
state_amplitude = df_detrend.groupby('state')['policy_detrended'].std().reset_index()
state_biennial = df.groupby('state')['biennial_legislature'].first().reset_index()
df_h2 = pd.merge(state_amplitude, state_biennial, on='state')

# Regress amplitude on biennial dummy (descriptive check)
import statsmodels.api as sm
X = sm.add_constant(df_h2['biennial_legislature'])
fit_h2 = sm.OLS(df_h2['amplitude'], X).fit()
print(f'\n=== H2 descriptive: Amplitude (detrended SD) vs. Biennial Legislature (De
print(fit_h2.summary().tables[1])

```

=== H2 descriptive: Amplitude (detrended SD) vs. Biennial Legislature (Delay Proxy)

===

=====

=====

|  | coef | std err | t | P> t | [0.025 | 0. |
|--|------|---------|---|------|--------|----|
|--|------|---------|---|------|--------|----|

975]

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----

|       |        |       |        |       |       |  |
|-------|--------|-------|--------|-------|-------|--|
| const | 0.4202 | 0.024 | 17.574 | 0.000 | 0.372 |  |
|-------|--------|-------|--------|-------|-------|--|

0.468

|                      |         |       |        |       |        |   |
|----------------------|---------|-------|--------|-------|--------|---|
| biennial_legislature | -0.1814 | 0.085 | -2.125 | 0.039 | -0.353 | - |
|----------------------|---------|-------|--------|-------|--------|---|

0.010

=====

=====